

Lightweight Proxy ~~EM~~Mamba Validation for Polyp Segmentation

~~EM~~Mamba: An Efficient Medical Segmentation Model With Edge Enhancement
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1. Executive Summary

This report documents the design, optimization, and empirical validation of a modified, CPU-friendly variant of the ~~EM~~Mamba architecture applied to the Kvasir-SEG dataset. To bypass the local hardware limitations of compiling the official CUDA-dependent `mamba_ssm` library, a proxy architecture was built using 1D depthwise convolutions to simulate sequence scanning.

Optimized under tight constraints (maximum 1-hour training time on a standard CPU), the pipeline was accelerated using modern training optimization methodologies (AdamW and OneCycleLR scheduling). Over a 10-epoch execution run using 880 training images, the model demonstrated distinct convergence behaviors, leaping from an initial trial validation Dice score of 0.26 to a peak validation Dice score of 0.5767 and a finalized test Dice score of 0.4284. Simultaneously, the model achieved a high-throughput CPU inference footprint of 0.075 seconds per image (~13.3 FPS).

2. Technical Architecture Comparison

The experimental pipeline swaps out the computationally intensive, hardware-bound elements of the original ~~EM~~Mamba paper with lightweight algorithmic substitutes while maintaining the overall structural macro-architecture.

Architectural Component	Original EM Mamba Framework	Lightweight Experimental Proxy	Impact of Modification
Sequence Encoder	Vision Mamba (Vim) Block with true Selective State Space Models (SSM)	SimpleMambaBlock using 1D Depthwise Convolutions (k=5)	Sacrifices Global Context: Swaps infinite receptive fields for local context windows, but drops custom CUDA dependencies to enable native CPU execution.
Edge Extraction	Edge-Aware Module (EAM) with dual Sobel kernels and neural gating	Dedicated EdgeAwareModule with registered Sobel tensors	Preserved: Retains structural edge extraction to explicitly map ambiguous boundary gradients.

Feature Upsampling	Efficient Fusion Bridge (EFB) with spatial/channel cross-attention	MultiScaleFusion with Bilinear Interpolation & 1 * 1 convolutions	Reduces Overhead: Replaces heavy cross-attention matrices with deterministic channel matching and concatenation blocks.
Boundary Modeling	Boundary Sensitive Decoder (BSD) using inverse-attention mechanisms	Convolutional decoder upsampling to original spatial resolution	Reduces Edge Focus: Decodes features linearly rather than forcing residual components to calculate mathematical target boundaries.

3. Experimental Setup & Environment

The experiment utilized a partition of the official Kvasir-SEG dataset. Training parameters were heavily adapted to align with a rigid time boundary while maximizing information processing per minute.

Parameter Type	Configuration Value	Operational Purpose / Notes
Computer Hardware	Central Processing Unit (CPU)	Standard hardware setting without specialized GPU acceleration.
Dataset Distribution	880 Train 60 Validation 60 Test	Reconfigured from original small-batch code to utilize the full volume of available entries.
Image Resolution	192 * 192 pixels	Downscaled from native resolutions to drastically reduce tensor size during matrix multiplication.
Batch Size	2	Set minimum to prevent out-of-memory (OOM) tracking on CPU RAM pools.
Base Learning Rate	$5.0 * 10^{-3}$ (Peak)	Elevated to encourage rapid feature assignment across limited epoch exposures.
Optimization Method	AdamW	Decouples weight decay from the gradient update step for cleaner regularized updates.
LR Scheduler	OneCycleLR	Forces cyclical escalation and decay to initiate super-convergence dynamics.
Total Constraints	10 Epochs max	Tuned directly to yield a target execution envelope under 60 minutes.

4. Training Dynamic Metrics Log

The following table tracks the complete performance history recorded across the 10 training epochs.

Epoch	Objective Loss	Validation Dice Score	Validation IoU Score	Training Insights & Dynamic States
1	1.0038	0.4615	0.3168	Initial discovery; structural edges begin to register.
2	1.0355	0.3594	0.2361	LR Escalation Phase: Learning rate rises, creating transient metric volatility.
3	1.0550	0.4002	0.2623	Loss tracking peaks; model undergoes heavy parameter correction.
4	1.0338	0.2238	0.1413	Lowest validation step; weights are highly unstable at maximum LR bounds.

5	0.9742	0.4344	0.2871	Cool-Down Phase Beginning: Learning rate begins to contract; stability returns.
6	0.9322	0.4880	0.3360	Metrics push past initial baselines; objective loss steadily tracks downward.
7	0.9039	0.5139	0.3607	Surpasses 0.50 Dice threshold; edge boundaries solidify across evaluation images.
8	0.8760	0.5298	0.3806	Convergence curves flatten out; precision optimization kicks in.
9	0.8360	0.5734	0.4201	Final optimization step; learning rate falls to near-zero boundaries.
10	0.8165	0.5767	0.4213	Optimal Performance: Minimum loss achieved. Weight snapshot saved.

5. Final Performance and Baseline Benchmarking

Upon completing the final epoch loop, the runtime generated the localized model state file (`best_emmamba_core.pth`). This checkpoint was evaluated against an independent, unseen testing block and benchmarked relative to contextual standards.

Metric Type	Lightweight Experimental Code Result	Target SOTA Baseline (Full Model)	System Implications & Analysis
Test Set Dice Score	0.4284	~0.9000+	Reflects expected limits of local 1D kernel windows versus true global sequence scanning.
Test Set IoU Score	0.3132	~0.8500+	Indicates that while the model locates the general presence of a polyp, boundaries remain coarse.
Inference Footprint	0.075 sec / img	Variable (Hardware Bound)	Highly optimized. Operates efficiently on a standard CPU, producing a fast, real-time feedback loop.
Artifact Generation	<code>best_emmamba_core.pth</code>	Complete weights	Confirms structural verification, validation loops, and weight saving actions functioned as designed.

6. Convergence Analysis & Core Findings

6.1. The OneCycleLR "V-Curve" Behavior

The metrics highlight a classic machine learning phenomenon associated with cyclical optimization schedules:

- **Epochs 2-4 (The Performance Dip):** As the OneCycleLR system forced the learning rate to its maximum peak (5.0×10^{-3}), the large updates destabilized validation metrics. This is why the Dice score briefly plummeted to 0.2238.

- Epochs 5-10 (The Super-Convergence Curve): As the scheduler cooled the learning rate down, the model effectively consolidated the structural patterns discovered during the high-velocity phases. It cleanly converged to its best validation state (0.5767) in the final step.

6.2. Structural Performance Boundaries

While the proxy model dramatically improved upon the initial prototype run (raising the Dice score from 0.26 to 0.42 on the final test), it cannot achieve true State-of-the-Art (SOTA) performance without global sequence modeling. A 1D depthwise convolution lacks the mathematical capacity to model long-range spatial context inside an image sequence. It acts as an effective structural validator, but a true selective scan model is required to resolve complex tissue camouflage.

7. Next Steps for Optimization

Phase 1: Scale Epochs (10 -> 50)

Phase 2: Transition to GPU (CUDA)

Phase 3: Native Mamba Blocks

To take this functional pipeline and systematically bridge the remaining performance gap toward true SOTA tracking, the following progression is recommended:

1. Extend Epoch Scale: Keep the current lightweight framework but extend execution from 10 to 50 epochs on a machine with a graphics processor to determine where the 1D convolution performance ceiling plateaus.
2. Transition to GPU Processing: Scale the input tensor configurations to larger resolutions (384 * 384) and expand the batch size to 16 using a native cuda environment.
3. Integrate Real Selective Scanning: Once working on a GPU machine, replace the SimpleMambaBlock depthwise convolution layer with the official Mamba selective scan module from the mamba_ssm package to unlock true global context learning.